

Physics 513 Midterm 2

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1. 1

In one of the homework problems, you studied the fake interacting field theory that is obtained by making a field redefinition of the free scalar field ϕ . Upon making a field redefinition you obtained interaction terms of various kinds. In this problem we will focus ONLY on the following two:

$$\mathcal{L}_{\text{int}} = -2\lambda\phi(\partial\phi)^2 + 8\lambda^2\phi^2(\partial\phi)^2. \quad (1)$$

What are the Feynman rules for the two vertices corresponding to these terms in the interaction Lagrangian. Write down and simplify the contributions of only these to the S-matrix for the process $\phi\phi \rightarrow \phi\phi$.

1.1. Lagrangian

$$\partial_\alpha\phi^2 = 2\phi\partial_\alpha\phi, \quad \partial_\alpha\phi^3 = 3\phi^2\partial_\alpha\phi \quad (2)$$

$$\begin{aligned} -2\lambda\phi(\partial\phi)^2 &= -2\lambda\eta^{\alpha\beta}\phi(\partial_\alpha\phi)(\partial_\beta\phi) = \\ &= -\lambda\eta^{\alpha\beta}(\partial_\alpha\phi^2)(\partial_\beta\phi) = \lambda\phi^2\eta^{\alpha\beta}\partial_\alpha\partial_\beta\phi = \lambda\phi^2\Box\phi \end{aligned} \quad (3)$$

$$\begin{aligned} 8\lambda^2\phi^2(\partial\phi)^2 &= 8\lambda^2\phi^2\eta^{\alpha\beta}(\partial_\alpha\phi)(\partial_\beta\phi) = \\ \frac{8}{3}\lambda^2\eta^{\alpha\beta}(\partial_\alpha\phi^3)(\partial_\beta\phi) &= -\frac{8}{3}\lambda^2\phi^3\eta^{\alpha\beta}\partial_\alpha\partial_\beta\phi = -\frac{8}{3}\lambda^2\phi^3\Box\phi \end{aligned} \quad (4)$$

$$\mathcal{L}_{\text{int}} = \lambda\phi^2\Box\phi - \frac{8}{3}\lambda^2\phi^3\Box\phi \quad (5)$$

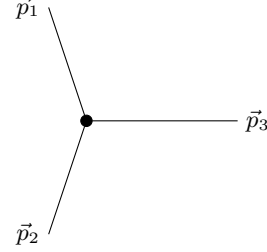


FIG. 1. The 3-way vertex.

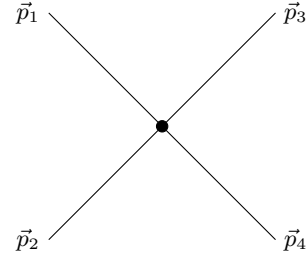


FIG. 2. The 4-way vertex.

1.2. Feynman Rules

The 3-way vertex (Fig. 1) Feynman rule is

$$\lambda\phi^2\Box\phi \Leftrightarrow -2i\lambda[(\vec{p}_1)^2 + (\vec{p}_2)^2 + (\vec{p}_3)^2], \quad (6)$$

where the momenta can be either entering or leaving the vertex. I use arrows for 4-vectors and bold face for 3-vectors.

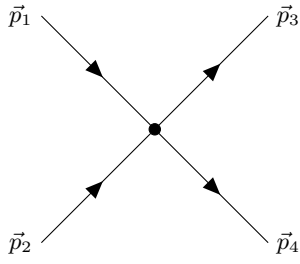
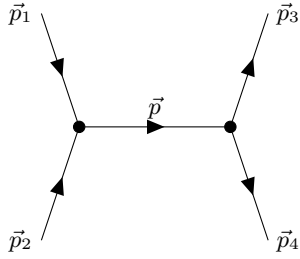
The 4-way vertex (Fig. 2) Feynman rule is

$$-\frac{8}{3}\lambda^2\phi^3\Box\phi \Leftrightarrow 16i\lambda^2[(\vec{p}_1)^2 + (\vec{p}_2)^2 + (\vec{p}_3)^2 + (\vec{p}_4)^2]. \quad (7)$$

1.3. Matrix Element

I work out the matrix element (nontrivial part of the scattering matrix) $\mathcal{M}$ to leading order in λ . Let the initial momenta be $\vec{p}_1$ and $\vec{p}_2$. Let the final momenta be $\vec{p}_3$ and $\vec{p}_4$. The total incoming and outgoing momenta (respectively) are

$$\vec{P} = \vec{p}_1 + \vec{p}_2, \quad \vec{K} = \vec{p}_3 + \vec{p}_4. \quad (8)$$

FIG. 3. Feynman diagram $\mathfrak{V}$.FIG. 4. $\mathfrak{V}[s]$

Because this is 2 to 2 scattering I use the Mandelstam variables

$$\begin{aligned} s &= (\vec{p}_1 + \vec{p}_2)^2 = (\vec{p}_3 + \vec{p}_4)^2, \\ t &= (\vec{p}_1 - \vec{p}_3)^2 = (\vec{p}_2 - \vec{p}_4)^2, \\ u &= (\vec{p}_1 - \vec{p}_4)^2 = (\vec{p}_2 - \vec{p}_3)^2. \end{aligned} \quad (9)$$

The matrix element is

$$\delta^{(4)}(\vec{P} - \vec{K}) i\mathcal{M} = \mathfrak{V} + \mathfrak{V}[s] + \mathfrak{V}[t] + \mathfrak{V}[u], \quad (10)$$

where $\mathfrak{V}$ (Fig. 3) is a 4-way diagram and the $\mathfrak{V}$ (Figs. 4-6) are 3-ways diagrams.

$$\mathfrak{V} = 16i\lambda^2 [(\vec{p}_1)^2 + (\vec{p}_2)^2 + (\vec{p}_3)^2 + (\vec{p}_4)^2] \delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4) \quad (11)$$

For any on-shell momentum: $\vec{p}^2 = m^2$.

$$\mathfrak{V} = 64i\lambda^2 m^2 \delta^{(4)}(\vec{P} - \vec{K}) \quad (12)$$

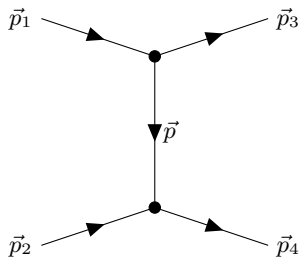
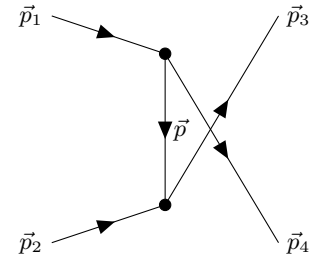
FIG. 5. $\mathfrak{V}[t]$ 

FIG. 6. Feynman diagram for $\mathfrak{V}[u]$. Notice that the crossing between $\vec{p}_3$ and $\vec{p}_4$ does not have a dot. That means it is not a vertex.

$$\mathfrak{V}[s] = (-2i\lambda)^2 i \int d^4p \frac{\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p})}{\vec{p}^2 - m^2 + i\epsilon} \delta^{(4)}(\vec{p} - \vec{p}_3 - \vec{p}_4) [(\vec{p}_1)^2 + (\vec{p}_2)^2 + \vec{p}^2][(\vec{p}_3)^2 + (\vec{p}_4)^2 + \vec{p}^2] \quad (13)$$

$$\mathfrak{V}[s] = -4\lambda^2 i \frac{(\vec{P}^2 + 2m^2)^2}{\vec{P}^2 - m^2} \delta^{(4)}(\vec{P} - \vec{K}) \quad (14)$$

$$\mathfrak{V}[s] = -4\lambda^2 i \frac{(s + 2m^2)^2}{s - m^2} \delta^{(4)}(\vec{P} - \vec{K}) \quad (15)$$

$$\mathfrak{V}[t] = (-2i\lambda)^2 i \int d^4p \frac{\delta^{(4)}(\vec{p}_1 - \vec{p}_3 - \vec{p})}{\vec{p}^2 - m^2 + i\epsilon} \delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_4) [(\vec{p}_1)^2 + (\vec{p}_3)^2 + \vec{p}^2][(\vec{p}_2)^2 + (\vec{p}_4)^2 + \vec{p}^2] \quad (16)$$

$$\mathfrak{V}[t] = -4\lambda^2 i \frac{(t + 2m^2)^2}{t - m^2} \delta^{(4)}(\vec{P} - \vec{K}) \quad (17)$$

$$\mathfrak{V}[u] = (-2i\lambda)^2 i \int d^4p \frac{\delta^{(4)}(\vec{p}_1 - \vec{p}_4 - \vec{p})}{\vec{p}^2 - m^2 + i\epsilon} \delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_3) [(\vec{p}_1)^2 + (\vec{p}_4)^2 + \vec{p}^2][(\vec{p}_2)^2 + (\vec{p}_3)^2 + \vec{p}^2] \quad (18)$$

$$\mathfrak{V}[u] = -4\lambda^2 i \frac{(u + 2m^2)^2}{u - m^2} \delta^{(4)}(\vec{P} - \vec{K}) \quad (19)$$

$$\boxed{\frac{\mathcal{M}}{4\lambda^2} = 16m^2 - \frac{(s + 2m^2)^2}{s - m^2} - \frac{(t + 2m^2)^2}{t - m^2} - \frac{(u + 2m^2)^2}{u - m^2}} \quad (20)$$

2. 2

Consider a particle (field operator ϕ) of mass m decaying into two particles (field operator χ) of mass μ . The interaction Lagrangian for this theory is given by $\mathcal{L}_{\text{int}} = g\phi(\partial_\mu\chi)^2$.

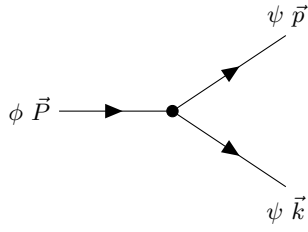


FIG. 7. Feynman diagram 7.

Find the total decay rate to leading non-trivial order in perturbation theory, assuming that $m > 2\mu$.

I don't like χ . So I relabel: $\chi \rightarrow \psi$.

$$\mathcal{L}_{\text{int}} = g \phi \eta^{\mu\nu} (\partial_\mu \psi) (\partial_\nu \psi) = g \phi (\partial \psi)^2 = \frac{2}{2!} g \phi (\partial \psi)^2 \quad (21)$$

The Feynman rule for this vertex is

$$\boxed{\mathcal{L}_{\text{int}} \Leftrightarrow -2ig(\pm \vec{p}) \cdot (\pm \vec{k})}, \quad (22)$$

where $\vec{p}$ and $\vec{k}$ correspond to ψ fields and $-/+$ is for entering/leaving the vertex.

The decay in this problem is $\vec{P}(\phi) \rightarrow \vec{p}(\psi)\vec{k}(\psi)$. The initial total 4-momentum is $\vec{P}$ and the final total 4-momentum is $\vec{K} = \vec{p} + \vec{k}$. The leading order matrix element for this process

$$\delta^{(4)}(\vec{P} - \vec{K}) i\mathcal{M} = \mathfrak{F}, \quad (23)$$

has one Feynman diagram 7 (Fig. 7).

$$\delta^{(4)}(\vec{P} - \vec{K}) i\mathcal{M} = -2ig \vec{k} \cdot \vec{p} \delta^{(4)}(\vec{P} - \vec{p} - \vec{k}) \quad (24)$$

$$\cancel{\delta^{(4)}(\vec{P} - \vec{K})} \mathcal{M} = -2g \vec{k} \cdot \vec{p} \cancel{\delta^{(4)}(\vec{P} - \vec{K})} \quad (25)$$

$$\boxed{\mathcal{M} = -2g \vec{k} \cdot \vec{p}} \quad (26)$$

I choose to work in the frame where the decaying particle is stationary

$$\vec{P} = (m, 0, 0, 0), \quad \vec{p} = (E, p, 0, 0), \quad \vec{k} = (E, -p, 0, 0), \quad (27)$$

where $p > 0$ and

$$E = \sqrt{\mu^2 + p^2}. \quad (28)$$

I get the differential decay rate from Schwartz Eq. (5.24)

$$d\Gamma = \frac{|\mathcal{M}|^2}{2m} d\Pi_{\text{LIPS}}, \quad (29)$$

where from Schwartz Eq. (5.21) I get

$$d\Pi_{\text{LIPS}} = \frac{d^3p d^3k}{16\pi^2 E_p E_k} \delta^{(4)}(\vec{P} - \vec{K}), \quad (30)$$

where $E_k = \sqrt{\mu^2 + k^2}$ and $E_p = \sqrt{\mu^2 + p^2}$. Because I am integrating over the 3-momenta $\mathbf{p}$ and $\mathbf{k}$ I cannot treat the energies E_k and E_p as the same here, even though I know that $E_k = E_p = E$. In the integral I use

$$\vec{p} = (E_p, \mathbf{p}), \quad \vec{k} = (E_k, \mathbf{k}). \quad (31)$$

$$32\pi^2 m d\Gamma = d^3p d^3k \frac{|\mathcal{M}|^2}{E_p E_k} \delta^{(4)}(\vec{P} - \vec{K}) \quad (32)$$

$$32\pi^2 m d\Gamma = d^3p d^3k \frac{|\mathcal{M}|^2 \delta^{(4)}(\vec{P} - \vec{p} - \vec{k})}{\sqrt{(\mu^2 + k^2)(\mu^2 + p^2)}} \quad (33)$$

$$32\pi^2 m d\Gamma = d^3p d^3k \frac{|-2g \vec{k} \cdot \vec{p}|^2 \delta^{(4)}(\vec{P} - \vec{p} - \vec{k})}{\sqrt{(\mu^2 + k^2)(\mu^2 + p^2)}} \quad (34)$$

$$\frac{8\pi^2 m}{g^2} d\Gamma = d^3p d^3k \frac{|\vec{k} \cdot \vec{p}|^2 \delta^{(4)}(\vec{P} - \vec{p} - \vec{k})}{\sqrt{(\mu^2 + k^2)(\mu^2 + p^2)}} \quad (35)$$

$$A = \frac{8\pi^2 m}{g^2} \quad (36)$$

$$A d\Gamma = d^3p d^3k \frac{|(E_k, \mathbf{k}) \cdot (E_p, \mathbf{p})|^2}{\sqrt{(\mu^2 + k^2)(\mu^2 + p^2)}} \delta^{(4)}(\vec{P} - \vec{p} - \vec{k}) \quad (37)$$

$$A d\Gamma = d^3p d^3k \frac{(E_k E_p - \mathbf{k} \cdot \mathbf{p})^2}{\sqrt{(\mu^2 + k^2)(\mu^2 + p^2)}} \delta^{(4)}(\vec{P} - \vec{p} - \vec{k}) \quad (38)$$

$$\mathbf{k} = -k \hat{\mathbf{x}}, \quad \mathbf{p} = p \hat{\mathbf{x}}, \quad p > 0, \quad k > 0 \quad (39)$$

$$A d\Gamma = d^3p d^3k \frac{[\sqrt{(\mu^2 + k^2)(\mu^2 + p^2)} + kp]^2}{\sqrt{(\mu^2 + k^2)(\mu^2 + p^2)}} \delta^{(4)}(\vec{P} - \vec{p} - \vec{k}) \quad (40)$$

$$A \Gamma = \int d^3p d^3k \frac{[\sqrt{(\mu^2 + k^2)(\mu^2 + p^2)} + kp]^2}{\sqrt{(\mu^2 + k^2)(\mu^2 + p^2)}} \delta^{(4)}(\vec{P} - \vec{p} - \vec{k}) \quad (41)$$

$$A \Gamma = \int d^3p d^3k \frac{[\sqrt{(\mu^2 + k^2)(\mu^2 + p^2)} + kp]^2}{\sqrt{(\mu^2 + k^2)(\mu^2 + p^2)}} \delta(m - E_p - E_k) \delta^{(3)}(\mathbf{P} - \mathbf{p} - \mathbf{k}) \quad (42)$$

$$\mathbf{P} = \mathbf{0} \quad (43)$$

$$f'(x) = -\frac{2x}{\sqrt{\mu^2 + x^2}} \quad (57)$$

$$A\Gamma = \int d^3p d^3k \frac{[\sqrt{(\mu^2 + k^2)(\mu^2 + p^2)} + kp]^2}{\sqrt{(\mu^2 + k^2)(\mu^2 + p^2)}} \delta\left(m - \sqrt{\mu^2 + p^2} - \sqrt{\mu^2 + k^2}\right) \delta^{(3)}(\mathbf{p} + \mathbf{k}) \quad (44)$$

$$f'(x_0) = -\frac{2\sqrt{m^2 - 4\mu^2}}{m} \implies |f'(x_0)| = \frac{2\sqrt{m^2 - 4\mu^2}}{m} \quad (58)$$

$$\frac{4\pi}{g^2} \sqrt{m^2 - 4\mu^2} \Gamma = \frac{m^2 - 4\mu^2}{4m^2} (m^2 - 2\mu^2)^2 \quad (59)$$

$$A\Gamma = \int d^3p \frac{[\sqrt{(\mu^2 + p^2)^2} + p^2]^2}{\sqrt{(\mu^2 + p^2)^2}} \delta\left(m - 2\sqrt{\mu^2 + p^2}\right) \quad (45)$$

$$\boxed{16\pi m^2 \Gamma = g^2 \sqrt{m^2 - 4\mu^2} (m^2 - 2\mu^2)^2} \quad (60)$$

$$A\Gamma = \int d^3p \frac{(\mu^2 + 2p^2)^2}{\mu^2 + p^2} \delta\left(m - 2\sqrt{\mu^2 + p^2}\right) \quad (46)$$

2.1. Correction

Because the final particles are identical I need to divide by $2!$.

$$\Gamma \rightarrow \frac{\Gamma}{2!} \quad (61)$$

$$\frac{8\pi^2 m}{g^2} \Gamma = \int_0^{4\pi} d\Omega \int_0^\infty dp p^2 \frac{(\mu^2 + 2p^2)^2}{\mu^2 + p^2} \delta\left(m - 2\sqrt{\mu^2 + p^2}\right) \quad (47)$$

$$\boxed{32\pi m^2 \Gamma = g^2 \sqrt{m^2 - 4\mu^2} (m^2 - 2\mu^2)^2} \quad (62)$$

$$\frac{2\pi m}{g^2} \Gamma = \int_0^\infty dx x^2 \frac{(\mu^2 + 2x^2)^2}{\mu^2 + x^2} \delta\left(m - 2\sqrt{\mu^2 + x^2}\right) \quad (48)$$

$$f(x) = m - 2\sqrt{\mu^2 + x^2} \quad (49)$$

$$\frac{2\pi m}{g^2} \Gamma = \int_0^\infty dx x^2 \frac{(\mu^2 + 2x^2)^2}{\mu^2 + x^2} \delta[f(x)] \quad (50)$$

The function $f(x)$ only has one positive root $x_0 \geq 0$

$$x_0 = \frac{\sqrt{m^2 - 4\mu^2}}{2}. \quad (51)$$

$$m > 2\mu \implies m^2 - 4\mu^2 > 0 \quad (52)$$

On the range $0 \leq x_0$

$$\delta[f(x)] = \frac{\delta(x - x_0)}{|f'(x_0)|}. \quad (53)$$

$$\frac{2\pi m}{g^2} |f'(x_0)| \Gamma = \int_0^\infty dx x^2 \frac{(\mu^2 + 2x^2)^2}{\mu^2 + x^2} \delta(x - x_0) \quad (54)$$

$$\frac{2\pi m}{g^2} |f'(x_0)| \Gamma = x_0^2 \frac{(\mu^2 + 2x_0^2)^2}{\mu^2 + x_0^2} \quad (55)$$

$$\frac{2\pi m}{g^2} |f'(x_0)| \Gamma = \frac{m^2 - 4\mu^2}{4m^2} (m^2 - 2\mu^2)^2 \quad (56)$$